

2.1 Vector-Valued Functions

1. Find the domain of the following vector functions.

a) $\mathbf{r}(t) = \left\langle \sqrt{16-t^2}, \ln(t+3), \frac{1}{1-t} \right\rangle.$

$\sqrt{16-t^2}$ has a domain of $[-4, 4]$.

$\ln(t+3)$ has a domain of $(-3, \infty)$.

$\frac{1}{1-t}$ has a domain of $(-\infty, 1) \cup (1, \infty)$.

Therefore, $\mathbf{r}(t)$ has a domain of $(-3, 1) \cup (1, 4]$.

b) $\mathbf{r}(t) = \frac{t-2}{t+2}\mathbf{i} + \sin(t)\mathbf{j} + \ln(9-t^2)\mathbf{k}$

$\frac{t-2}{t+2}$ has a domain of $(-\infty, -2) \cup (-2, \infty)$.

$\sin(t)$ has a domain of $(-\infty, \infty)$.

$\ln(9-t^2)$ has a domain of $(-3, 3)$.

Therefore, $\mathbf{r}(t)$ has a domain of $(-3, -2) \cup (2, 3)$.

c) $\mathbf{r}(t) = \left\langle \frac{1}{t-2}, \sqrt{t+3}, \cos(t) \right\rangle$

$\frac{1}{t-2}$ has a domain of $(-\infty, 2) \cup (2, \infty)$.

$\sqrt{t+3}$ has a domain of $[-3, \infty)$.

$\cos(t)$ has a domain of $(-\infty, \infty)$.

Therefore, $\mathbf{r}(t)$ has a domain of $[-3, 2) \cup (2, \infty)$.

d) $\mathbf{r}(t) = \left\langle \ln(t^2-4), \frac{1}{\sqrt{t}}, t^3 \right\rangle$

$\ln(t^2-4)$ has a domain of $(-\infty, -2) \cup (2, \infty)$.

$\frac{1}{\sqrt{t}}$ has a domain of $(0, \infty)$.

t^3 has a domain of $(-\infty, \infty)$.

Therefore, $\mathbf{r}(t)$ has a domain of $(2, \infty)$.

e) $\mathbf{r}(t) = \arcsin(2t-1)\mathbf{i} + \sqrt{t}\mathbf{j} + e^{-t}\mathbf{k}$

For $\arcsin(2t-1)$ we need $-1 \leq 2t-1 \leq 1 \rightarrow 0 \leq t \leq 1$. Therefore, $\arcsin(2t-1)$ has a domain of $[0,1]$.

$\sqrt{t}$ has a domain of $[0, \infty)$.

e^{-t} has a domain of $(-\infty, \infty)$.

Therefore, $\mathbf{r}(t)$ has a domain of $[0,1]$.

2. Find any three different parameterizations of the graph $y = x^2$ on the interval $1 \leq x \leq 4$.

1) If $x = t$, then $y = t^2$ and $1 \leq t \leq 4$ so we have $\mathbf{r}(t) = \langle t, t^2 \rangle$ with $1 \leq t \leq 4$.

2) If $x = 2t$, then $y = (2t)^2 = 4t^2$ and $1 \leq 2t \leq 4$ so we have $\mathbf{r}(t) = \langle 2t, 4t^2 \rangle$ with $\frac{1}{2} \leq t \leq 2$.

3) If $x = t+1$, then $y = (t+1)^2$ and $1 \leq t+1 \leq 4$ so we have $\mathbf{r}(t) = \langle t+1, (t+1)^2 \rangle$ with $0 \leq t \leq 3$.

3. Find any three different parameterization of the graph of $y = \ln(x^2) + 2x$ on the interval $2 \leq x \leq 5$

1) If $x = t$, then $y = \ln(t^2) + 2t$ and $2 \leq t \leq 5$ so we have $\mathbf{r}(t) = \langle t, \ln(t^2) + 2t \rangle$ with $2 \leq t \leq 5$.

2) If $x = \sqrt{t}$, then $\ln(t) + 2\sqrt{t}$ and $2 \leq \sqrt{t} \leq 5$ so we have $\mathbf{r}(t) = \langle \sqrt{t}, \ln(t) + 2\sqrt{t} \rangle$ with $4 \leq t \leq 25$.

3) If $x = \frac{t}{2}$, then $y = \ln\left(\left(\frac{t}{2}\right)^2\right) + 2\left(\frac{t}{2}\right) = \ln\left(\frac{t^2}{4}\right) + t$ and $2 \leq \frac{t}{2} \leq 5$ so we have $\mathbf{r}(t) = \left\langle \frac{t}{2}, \ln\left(\frac{t^2}{4}\right) + t \right\rangle$ with $4 \leq t \leq 10$.

4. Find any three different parameterization for a single tracing of the graph of $x^2 + y^2 = 4$.

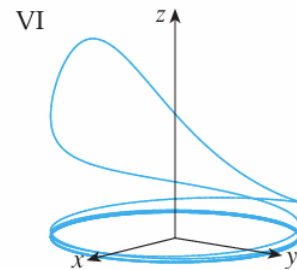
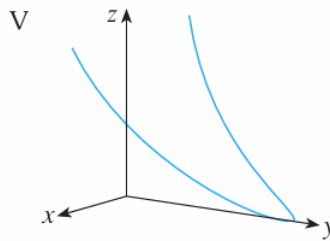
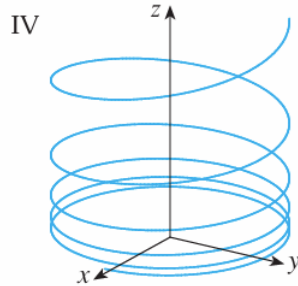
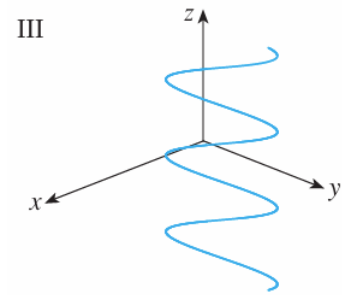
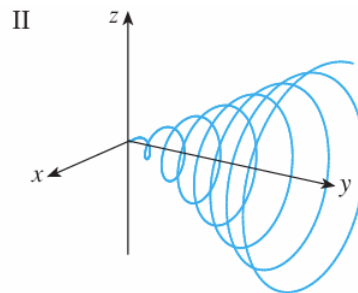
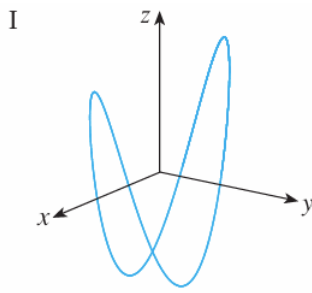
This is a circle of radius 2.

1) If $x = 2\cos(t)$, then $y = 2\sin(t)$ and $0 \leq t \leq 2\pi$ so we have $\mathbf{r}(t) = \langle 2\cos(t), 2\sin(t) \rangle$ with $0 \leq t \leq 2\pi$.

2) If $x = 2\sin(t)$, then $y = 2\cos(t)$ and $0 \leq t \leq 2\pi$ so we have $\mathbf{r}(t) = \langle 2\sin(t), 2\cos(t) \rangle$ with $0 \leq t \leq 2\pi$.

3) If $x = 2\cos(2t)$, then $y = 2\sin(2t)$ and $0 \leq t \leq 2\pi$ so we have $\mathbf{r}(t) = \langle 2\cos(2t), 2\sin(2t) \rangle$ with $0 \leq t \leq \pi$.

5. Match the parametric equations with the graphs below. Give reasons for your choices.



a) $\mathbf{r}(t) = \langle t \cos(t), t, t \sin(t) \rangle$ for $t \geq 0$.

In the x and z direction we have a circular shape (because of the sine and cosine) with an increasing radius (which comes from the coefficient of the t). Note that the radii of the circles increases as the y -coordinate increases.

Therefore, this vector function goes with graph II.

b) $\mathbf{r}(t) = \left\langle \cos(t), \sin(t), \frac{1}{1+t^2} \right\rangle$

In the x and y direction we have a circular shape (because of the sine and cosine) with a fixed radius of 1 (because of the coefficient of 1). The z -coordinate decreases (quickly and then more slowly) as t increases.

Therefore, this vector function goes with graph IV.

c) $\mathbf{r}(t) = \left\langle t, \frac{1}{1+t^2}, t^2 \right\rangle$

In the x and z direction we have a parabola shape. The y -coordinate decreases (quickly and then more slowly) as t increases.

Therefore, this vector function goes with graph V.

d) $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + \cos(2t)\mathbf{k}$

In the x and y direction we have a circular shape (because of the sine and cosine) with a fixed radius of 1 (because of the coefficient of 1). The z -coordinate also cycles between -1 and 1 but at a quicker rate than the x and y directions can form a circle. Specifically for every half circle sketched in the x and y directions the z coordinate fully oscillates between -1 and 1.

Therefore, this vector function goes with graph I.

e) $\mathbf{r}(t) = \cos(8t)\mathbf{i} + \sin(8t)\mathbf{j} + e^{0.8t}\mathbf{k}$ for $t \geq 0$.

In the x and y direction we have a circular shape (because of the sine and cosine) with a fixed radius of 1 (because of the coefficient of 1). The z -coordinate is always positive and increases (slowly at first and then more quickly).

Therefore, this vector function goes with graph VI.

f) $\mathbf{r}(t) = \cos^2(t)\mathbf{i} + \sin^2(t)\mathbf{j} + t\mathbf{k}$

In the x and y direction we have a values that oscillate between 0 and 1 (due to the sine and cosine being squared with leading coefficients of 1). The z -coordinate is always positive and increases at a linear rate.

Therefore, this vector function goes with graph III.

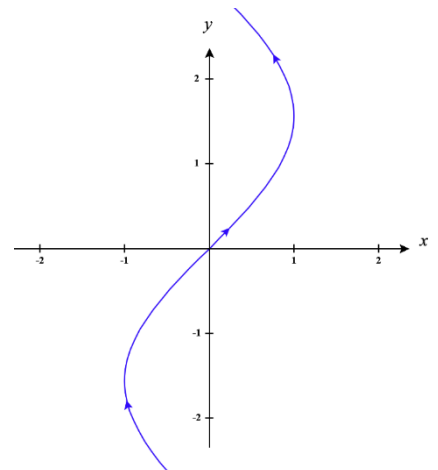
6. True or False: The functions $\mathbf{r}(t) = \langle \cos(3t), \sin(3t) \rangle$ and $\mathbf{s}(t) = \langle 3\cos(t), 3\sin(t) \rangle$ generate the same curve.

False. By simply sketching these curves we can see that they are not the same.

7. Sketch the curve with the given vector equation. Indicate with an arrow the direction in which t increases.

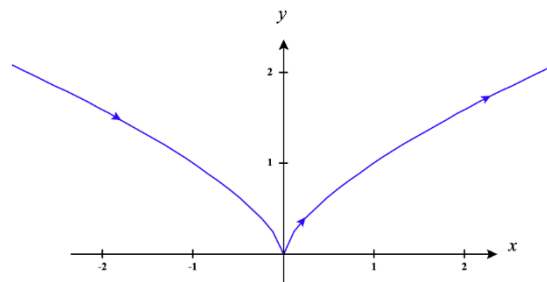
a) $\mathbf{r}(t) = \langle \sin(t), t \rangle$

Note that $x = \sin(t)$ & $y = t \rightarrow x = \sin(y)$.



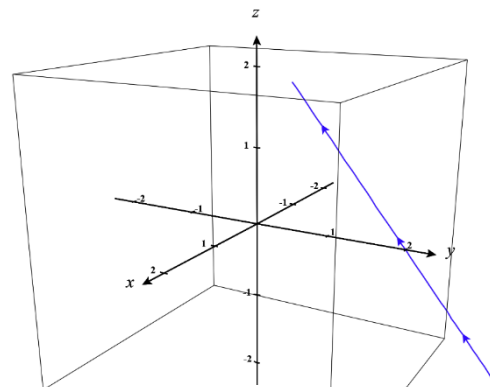
b) $\mathbf{r}(t) = \langle t^3, t^2 \rangle$

Note that $x = t^3$ & $y = t^2 \rightarrow y = x^{2/3}$.



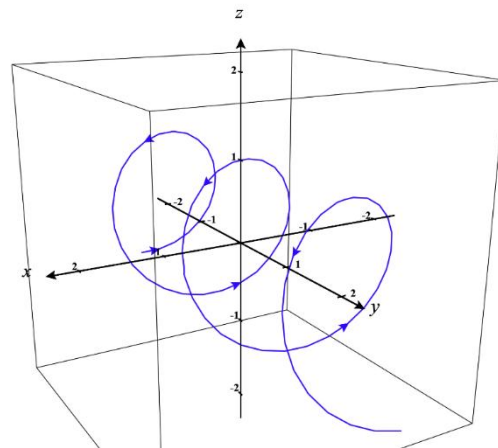
c) $\mathbf{r}(t) = \langle t, 2-t, 2t \rangle$

We can see right away this is the equation of a line from chapter 1. When $t = 0$ the function is at the point $(0, 2, 0)$ and when $t = 1$ the function is at the point $(1, 1, 2)$.



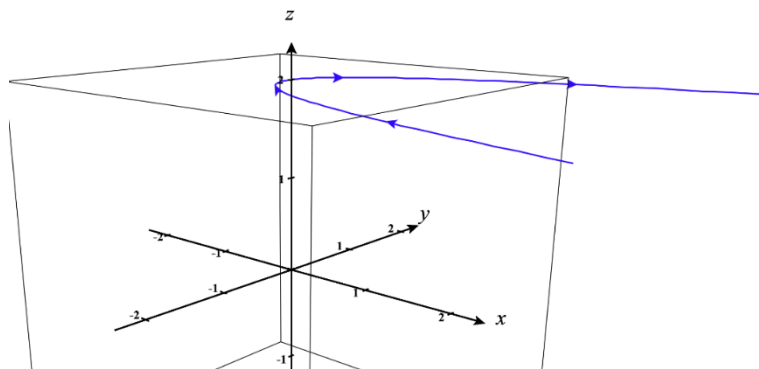
d) $\mathbf{r}(t) = \langle \sin(\pi t), t, \cos(\pi t) \rangle$

In the x and z directions we have circles of radius 1. The y -coordinate increases without bound as the circles are sketched.



e) $\mathbf{r}(t) = t^2\mathbf{i} + t\mathbf{j} + 2\mathbf{k}$

In the x and y directions we have a sideways parabola. The z -coordinate is fixed at 2.



f) $\mathbf{r}(t) = \sin(t)\mathbf{i} + \cos(t)\mathbf{j} + 2\mathbf{k}$

In the x and y directions we have circles of radius 1. The z -coordinate is fixed at 2.

